Exercise: OLTP Using LINQPad

Marking Rubric (1 Mark - Pass/Fail) No Partial Marks

Weight	Criteria
1	**Proficient** – Requirement is met
0	**Incomplete** – Requirement is not met, contains major errors, or is missing significant portions

Grading Rubric

Each test is worth **1 mark**. **No partial marks.**

Employee Tests (6 Marks)

Test Name	Criteria	Marks
`Valid_EmptyLastNameAndPhone`	Ensures that searching with an empty last name and phone number returns no results.	1
`Valid_PartialPhoneNumber`	Ensures that searching with a partial phone number returns no results.	1
`Valid_ActualPhoneNumber`	Ensures that searching with an actual phone number returns the correct employee.	1
`Valid_PartialLastName_Return_Four_Employees`	Ensures that searching with a partial last name returns four specific employees.	1
`Valid_EmployeeData`	Ensures that the returned employee data matches the expected values.	1

Test Name	Criteria	Marks
`Valid_TerritoryData`	Ensures that the employee's territory data is accurate.	1

Product Tests (3 Marks)

Test Name	Criteria	Marks
`Valid_ProductID_Zero`	Ensures that searching for a product with an invalid ID returns no results.	1
`Valid_Product_Removed`	Ensures that removed products do not appear in the results.	1
`Valid_Product`	Ensures that a valid product ID returns the correct product details.	1

Order Tests (16 Marks)

Test Name	Criteria	Marks
`Valid_OrderID_Zero`	Ensures that searching for an order with an invalid ID returns no results.	1
`Valid_Order_Removed`	Ensures that removed orders do not appear in the results.	1
`Valid_Order_OrderID`	Ensures that a valid order ID returns the correct order details.	1
Valid_AddEditOrder	Ensures that adding or editing an order follows business rules, validation, and saves correctly.Max 13 Marks.	n/a

Breakdown for `Valid_AddEditOrder` (13 Additional Marks)

Validation Checks (5 Marks)

Rule	Description	Marks
Order cannot be null	Ensures an order is supplied.	1

Rule	Description	Marks
Customer & Employee ID must be provided	Ensures that the customer & employee ID is required.	1
Order must have at least one order line	Ensures that an order contains order details.	1
Each order line must have a product	Ensures that every order detail has a valid product ID.	1
Product cannot be duplicated in multiple order lines	Ensures that the same product does not appear multiple times.	1

Business Rule Checks (3 Marks)

Rule	Description	Marks
Unit price cannot be negative	Ensures the unit price for an order line is zero or greater.	1
Quantity must be at least one	Ensures that each order line has a quantity greater than zero.	1
Order date must be created for new orders only.	Ensure that the that new order at set to "Today" date.	1

Data Integrity Checks (5 Marks)

Type	Rule	Description	Marks
Adding a New Order	Order must have a valid OrderID	Ensures the order is successfully created in the system.	1
Adding a New Order	Order must have the correct data added for it (Customer, Employee, etc.)	Verifies that the provided information is correctly added to the order	1
Editing an Existing Order	Existing order data must be updated correctly	Ensures changes to the order, such as employee ID updates, are properly saved.	1

Type	Rule	Description	Marks
Editing an Existing Order	Updated order must retain correct subtotal and tax calculations	Ensures the recalculated values reflect any changes made to the order.	1
Stock Management	Units in stock must be updated correctly	Ensures that product stock levels adjust based on added or modified order quantities.	1

Total Marks: 25

- **All tests must pass to receive marks.**
- **If a test fails, no marks are awarded for that test.**

Penalty: Up to **-3 marks** for insufficient commits, commented-out code, or improper file naming.

Important: First Execution Policy

Upon the **first execution** of your submitted code by the instructor, if it **fails to run**, you will receive a **zero**.

Submission Requirements

*All queries must use LINQ Method Syntax** (Query Syntax is **not allowed**).
*Create all methods and view models in the designated area** in the `.linq` file (**see instructions below**).
*Save all queries in a single file** following this naming convention:

```
Take-home 3 - Your Name.linq
```

Academic Integrity Warning

DO NOT use AI-generated content (e.g., ChatGPT, Meta.AI, etc.).
*Using AI assistance may result in academic discipline.**

Overview

This exercise is based on the **WestWind-2025-Jan-TH3** database.
You are provided with a **starter LINQPad (`.linq`) file**.

Task Overview

To enhance customer satisfaction and streamline operations, you are required to **implement key system features** using **LINQ queries**.

The **UI development** will be handled by another team **after** your queries are complete.

🔹 Required Features

1. **Search for Employees** (by **Last Name** and/or **Phone Number**)
   - *Returns**: A **list** of employees and their assigned territories (`EmployeeView` & `TerritoryView`).
2. **Retrieve an Existing Product** (by `ProductID`)

- *Returns**: A **single** product (`ProductView`).

3. **Retrieve an Existing Order** (by `OrderID`)

- *Returns**: A **single** order (`OrderView`) along with a **list** of order details (`OrderDetailView`).

4. **Create/Edit an Order** (using `OrderView`)

- *Returns**: A **newly created or updated** order (`OrderView`) along with a **list** of order details (`OrderDetailView`).

These features are **critical** to maintaining an efficient and user-friendly system.
Your task is to **implement the required functionalities** using **LINQ queries**.
The UI will be developed by another team **after** you complete your queries.

Required Method Names

You **must** use the following method names:

- `GetEmployees`
- `GetProduct`
- `GetOrder`
- `AddEditOrder`

Required View Models & Properties

Ensure that each property has the correct data type when implementing your models.

EmployeeView

Represents employee details. Each employee may have multiple assigned territories. Order By last name

List<EmployeeView> (4 items)

EmployeeID	Title	FullName	Phone	Country	ReportTo	RemoveFromViewFlag	SalesTerritories
5	Sales Manager	Steven Buchanan	(71) 555-4848	UK	Andrew Fuller	False	
8	Inside Sales Coordinator	Laura Callahan	(206) 555-1189	USA	Andrew Fuller	False	
4	Sales Representative	Margaret Peacock	(206) 555-8122	USA	Andrew Fuller	False	
6	Sales Representative	Michael Suyama	(71) 555-7773	UK	Steven Buchanan	False	

List<TerritoryView> (7 items)

TerritoryID	TerritoryDescription	RegionDescription	RemoveFromViewFlag
08837	Edison	Eastern	False
14450	Fairport	Eastern	False
11747	Mellvile	Eastern	False
07960	Morristown	Eastern	False
10019	New York	Eastern	False
10038	New York	Eastern	False
02903	Providence	Eastern	False

List<TerritoryView> (4 items)

TerritoryID	TerritoryDescription	RegionDescription	RemoveFromViewFlag
44122	Beachwood	Northern	False
45839	Findlay	Northern	False
19428	Philadelphia	Northern	False
53404	Racine	Northern	False

List<TerritoryView> (3 items)

TerritoryID	TerritoryDescription	RegionDescription	RemoveFromViewFlag
27511	Cary	Eastern	False
27403	Greensboro	Eastern	False
20852	Rockville	Eastern	False

List<TerritoryView> (5 items)

TerritoryID	TerritoryDescription	RegionDescription	RemoveFromViewFlag
98004	Bellevue	Western	False
85014	Phoenix	Western	False
98052	Redmond	Western	False
85251	Scottsdale	Western	False
98104	Seattle	Western	False

Fields:

- EmployeeID
- Title
- FullName
- Phone
- Country
- ReportTo
- RemoveFromViewFlag
- SalesTerritories (A list of TerritoryView)

TerritoryView

Represents a sales territory. Order By territory description

Fields:

- TerritoryID
- TerritoryDescription
- RegionDescription
- RemoveFromViewFlag

ProductView

Represents product details.

▲ ProductView •••	
UserQuery+TakeHome3+ProductView	
ProductID	1
ProductName	Chai
SupplierID	1
CategoryID	1
QuantityPerUnit	10 boxes x 20 bags
UnitPrice	18.0000
UnitsInStock	1000
UnitsOnOrder	0
ReorderLevel	10
Discontinued	False
RemoveFromViewFlag	False

Fields:

```
- ProductID
- ProductName
- SupplierID
- CategoryID
- QuantityPerUnit
- UnitPrice
- UnitsInStock
- UnitsOnOrder
- ReorderLevel
- Discontinued
- RemoveFromViewFlag
```

OrderView

Represents an order. Each order may contain multiple **order details**.

OrderView

UserQuery+TakeHome3+OrderView

OrderID	11079
CustomerID	ALFKI
CompanyName	Alfreds Futterkiste
EmployeeID	2
EmployeeName	Andrew Fuller
OrderDate	2/23/2025 9:06:51 AM
RequiredDate	2/28/2025 9:06:51 AM
ShippedDate	2/25/2025 9:06:51 AM
ShipVia	2
Freight	12.5500
ShipName	UPS
ShipAddress	Side of the Road
ShipCity	Nowhere
ShipRegion	Up There
ShipPostalCode	T5Q1X2
ShipCountry	Canada
OrderDetails	List<OrderDetailView> (2 items)
SubTotal	2714.2000
Tax	135.710000
RemoveFromViewFlag	False

OrderDetailID	OrderID	ProductID	Name	UnitPrice	Quantity	Discount	UnitsInStock	RemoveFromViewFlag
2159	11079	4	Chef Anton's Cajun Seasoning	19.0000	17	0.13	983	False
2160	11079	5	Chef Anton's Gumbo Mix	21.3500	112	0	888	False
				40.3500	129	0.13	1871	

Fields:

- OrderID
- CustomerID
- CompanyName
- EmployeeID
- EmployeeName
- OrderDate
- RequiredDate
- ShippedDate
- ShipVia
- Freight
- ShipName
- ShipAddress
- ShipCity
- ShipRegion
- ShipPostalCode
- ShipCountry
- OrderDetails (A list of OrderDetailView)
- SubTotal
- Tax
- RemoveFromViewFlag

OrderDetailView

Represents an order detail, which includes product information and pricing. Order By name

Fields:

```
- OrderDetailID
- OrderID
- ProductID
- Name
- UnitPrice
- Quantity
- Discount
- UnitsInStock
- RemoveFromViewFlag
```

Business Rules for Methods (Reference the test for exact Error Messages)

🔹 `GetEmployees` Method (Search for Employees)

1. **At least one input is required** – Either `lastName` or `phoneNumber` must be provided.
   - **Error Message Must Contain**: `"A last name or phone number is required."`
2. **Phone number must be an exact match** – It cannot be a partial match.
3. **Only active employees are retrieved**

🔹 `GetProduct` Method (Retrieve a Product by `ProductID`)

1. **Product ID is required** – If `ProductID` is not valid, an Error is provided.
   - **Error Message Must Contain**: `"Please provide a product id."`
2. **Only active products are retrieved**

🔹 `GetOrder` Method (Retrieve an Order by `OrderID`)

1. **Order ID is required** – If `OrderID` is not valid, an Error is provided.
   - **Error Message Must Contain**: `"Please provide a invoice id."`
2. **Only active orders are retrieved** - If no order is found, an Error is provided.
   - **Error Message Must Contain**: `"No order was found with ID: {orderID}"`
3. **Subtotal and Tax calculations** –
4. **Only active order details are included**

🔹 `AddEditOrder` Method (Create/Edit an Order)

General Order Rules

1. **Order cannot be null** – If `orderView` is `null`, an Error is provided.
   - **Error Message Must Contain**: `"No order was supplied."`
2. **Customer ID is required** – The order must be associated with a valid customer.
   - **Error Message Must Contain**: `"Customer is required."`
3. **Employee ID is required** – An employee must be assigned to the order.
   - **Error Message Must Contain**: `"Employee is required."`
4. **At least one order detail is required** – The order must contain at least one product.
   - **Error Message Must Contain**: `"Order details are required."`

Order Line Validation

5. **Each order detail must have a valid product** – `ProductID` is not valid.
   - **Error Message Must Contain**: `"Missing product ID"`
6. **Product price cannot be negative** – `UnitPrice` must be greater than or equal to `0`.
   - **Error Message Must Contain**: `has a price that is less than zero`
     - i.e: `Product 'XBox Controller' has a price that is less than zero`
7. **Product quantity must be at least 1** – `Quantity` must be greater than or equal to `1`.
   - **Error Message Must Contain**: `has a quantity that is less than one`
     - i.e: `Product 'XBox Controller' has a quantity that is less than one`
8. **Products cannot be duplicated** – Each product can only appear once per order.
   - **Error Message Must Contain**: `can only be added to the order lines once`

- i.e: `Product 'XBox Controller' can only be added to the order lines once`

Order Processing

9. **New orders are assigned the current date** – `OrderDate = DateTime.Now` for new orders.
10. **New order details update stock levels** – If an order detail is newly added and not removed from view, `UnitsInStock` is reduced.

Final Validation and Save

11. **All errors are checked before saving** – If there are errors, changes are **rolled back**, and an `AggregateException` is thrown.
12. **Only one `SaveChanges()` call is allowed** – All updates must be completed before saving to maintain data integrity.

Important Information

Important Guidelines

- **DO NOT** modify any code other than the methods & view models that you are creating. You may update the "Warning Level" found in the driver.
  - Any other modifications, including removal of test classes, will result in a **zero (0)** for "Take Home #3".
- **Test methods have already been created.**
  - You are only required to implement the methods listed in the requirements document.
- **Your code must be written inside the `TakeHome3` class:**
  - Implement your methods within:

```
#region Take Home #3 Methods

#endregion
```

  - Create your view models within:

```
#region View Models

#endregion
```

- **You may update and change the test variables in the main driver.**

Development Instructions

- **Start by creating your view models.**
- **Stub out your methods based on the test method naming pattern.**
  - If the code behind method is `GetCustomer(xxx)` , your method should be `GetCustomer(xxx)` .
  - Add `return result;` to ensure your method has a return value.
- **Follow an ordered approach:**
  - Complete each method before moving on to the next, especially if later methods depend on previous ones.

Successful Run

```
*********************************************
***********  Employee Tests (6)  ************
*********************************************
Pass - Empty Last Name and Phone
Pass - Partial Phone Number Returned No Matches
Pass - Exact Phone Number Returned Correct Match
Pass - Partial Last Name Search 'a' Returned the 4 correct Employees
Pass - Expected Employee 'Anne Dodsworth' returned with correct values
Pass - Expected Employee 'Anne Dodsworth' returned with correct Territories in the correct order
****************************************
***********  Product Tests (3)  ***********
****************************************
Pass - Product ID '0' not accepted
Pass - Product ID '3' correctly returns no results
Pass - Product ID '1' correctly returns with correct values
*********************************************
***********  Get Order Tests (4)  ***********
*********************************************
Pass - Order ID '0' not accepted
Pass - Order ID '10249' correctly returns no results
Pass - Order ID '10248' correctly returns with correct values
Pass - Order '10248' returned the correct number of order detail records and the correct records
**************************************************
***********  Add/Edit Order Tests (13)  ***********
**************************************************
Pass - Correct Error Message returned when Order supplied is null
Pass - Correct Error Messages returned when Order supplied has no EmployeeID or CustomerID
Pass - Correct Error Message returned when Order supplied has no OrderDetails
Pass - Correct Error Message returned when Order supplied has OrderDetail with no Product ID
Pass - Correct Error Message returned when Order supplied has duplicate OrderDetails
Pass - Correct Error Message returned when OrderDetail has negative price
Pass - Correct Error Message returned when OrderDetail has a quantity of 0
Pass - New Order has a valid OrderID
Pass - New Order was provided a valid Date
Pass 1/2 - New Order added with all correct information
Pass 2/2 - New Order Added with correct Order Detail information
Pass - Edit Order correctly updated
Pass - Edit Order Subtotal and Tax correctly updated
Pass - Edit Order Inventory Levels correctly updated
```